

Software Requirements Specification (SRS)

Sentiment Analysis Web Application

1. Introduction

Purpose: Develop a web-based Sentiment Analysis system that classifies text into Positive, Negative, or Neutral categories using Naive Bayes or LSTM model. Built using Python Flask and deployed on Render cloud platform.

2. Scope

The system allows users to input text, preprocess it, predict sentiment using ML/DL model, and display results via web interface.

3. Overall Description

The system consists of frontend (HTML/CSS), backend (Flask), ML/DL model (Naive Bayes or LSTM), and cloud deployment (Render).

4. Functional Requirements

- User text input submission - Text preprocessing (lowercasing, stopword removal, tokenization) - Feature extraction (TF-IDF or Tokenization & Padding) - Sentiment classification - Display of results with optional confidence score

5. Non-Functional Requirements

- Response time under 3 seconds - Secure input validation - Scalable cloud deployment - 99% uptime reliability

6. System Architecture

User → Web Browser → Flask Backend → ML/DL Model → Prediction → Browser

7. Data Requirements

Dataset in CSV format with columns: Text, Sentiment. Example datasets: Movie Reviews, Twitter Sentiment datasets.

8. Deployment Requirements

- Push project to GitHub - Connect repository to Render - Build command: pip install -r requirements.txt - Start command: gunicorn app:app

9. Future Enhancements

- Multi-language support - Sentiment visualization charts - User authentication - API integration support

10. Constraints

- Limited free-tier cloud resources - LSTM requires more memory than Naive Bayes